Patrick Mockridge

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PROFILE AND METHOD

Process safety taught me that a plausible number is not a safe number — so every result here is either proved or carries its limits. Process safety engineer, 2006–2015; chartered 2013; independent protocol and verification work since 2016.

The method: specify the failure mode before the architecture; prove what can be proved; contain what cannot; record provenance with every result. DarkWow is containment; rnode-rust is continuity; Azoth is provenance; Mudra is verification in a familiar form.

SELECTED WORK

DarkWow
Designed for private organisations on a public chain, without an administrator or a public voter roll

2026– · Research prototype; Lean-verified circuits; HAZOP-reviewed

rnode-rust
A Rust node for the ρ-calculus: native object capabilities and machine-checked laws

2026 · Pre-release

Azoth
A process calculation library where every number carries its provenance

2026 · v0.1.x, not for design work

Mudra
Prove a calculation was done correctly without revealing the spreadsheet

2026 · Research build; v1.0.x

Records: patrickmockridge.github.io/?tab=darkwow · /?tab=rnode-rust · /?tab=azoth · /?tab=mudra

CHRONOLOGY

PERIOD	WORK
2016–present	Independent work: protocol software, formal verification, numerical systems. Current records: DarkWow, rnode-rust, Azoth, Mudra.
2011–2014	Bitcoin since 2011; London Ethereum community, 2014.
2006–2015	Chemical engineer, process safety; UK and North Africa. Chartered Professional Chemical Engineer, 2013.
2002–2006	MEng Chemical Engineering, University of Edinburgh.

RESEARCH AND WRITING

UNISS / Ingenesist; Romeo–Juliet framework. Selected writing.